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A Comparative Analysis of Machine Learning Models for Intrusion Detection using the CSE-CIC-IDS2018 Dataset

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Abstract*—With the rise in cyber threats, protecting network systems has never been more critical. This study presents a comparative analysis of machine learning models for enhancing Network Intrusion Detection Systems (NIDS) using the CSE-CIC-IDS2018 dataset. In order to improve model reliability, class imbalance issues were addressed using undersampling technique s, ensuring a balanced dataset for training. Five machine learning models—Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Logistic Regression, and XGBoost—were evaluated using standard performance metrics. Among these, XGBoost achieved the highest scores with 96% accuracy, 94% precision, 98% recall, and 96% F1. These results highlight XGBoost's exceptional capacity for identifying intrusions with high precision, making it a strong candidate for practical cybersecurity deployment. Overall, the findings underscore the value of advanced machine learning techniques in developing reliable and adaptive intrusion detection systems for modern network environments.

***Keywords*—network intrusion detection system, cybersecurity, machine learning models, CSE-CIC-IDS2018, XGBoost**

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I. INTRODUCTION

In today's hyper-connected digital landscape, cybersecurity has become a critical priority. With cybercriminals increasingly targeting enterprises, government institutions, and essential infrastructure, the impact of cyberattacks extends beyond financial losses to include operational disruptions, data breaches, and reputational damage [1], [2]. Global cybercrime damages are projected to exceed $10.5 trillion annually by 2025, underscoring the urgent need for robust and adaptive defense mechanisms [3]. For example, the average cost of a data breach in 2023 was approximately $4.35 million, affecting both businesses and public institutions [4]. Organizations are turning to Network Intrusion Detection Systems (NIDS) to combat these threats, which act as digital sentinels by monitoring network traffic for signs of malicious activity [5]. Traditional NIDS models, based on static rule sets or early machine learning approaches such as decision trees and support vector machines (SVM), have become less effective against today's sophisticated threats. As attackers adopt more complex strategies—including zero-day exploits, polymorphic malware, and Advanced Persistent Threats (APTs)—NIDS must evolve accordingly [6]. Recent machine learning (ML) developmen ts offer promising tools for improving intrusion detection accuracy and adaptability. However, the success of ML-based NIDS relies heavily on two key factors: the quality of the dataset used for training and the effectiveness of preprocessing techniques [7]. While influential in early research, older datasets such as KDD Cup'99 and NSL-KDD no longer reflect the diversity and complexity of modern cyberattacks. These datasets often contain redundant or outdated attack patterns, limiting their usefulness for real-world applications [8], [9]. Researchers have shifted toward more comprehensive datasets like CSE-CIC-IDS2018, developed by the University of New Brunswick. This dataset includes over 16 million records and covers many modern attack types such as DDoS, brute-force, botnets, and infiltration attacks, offering a realistic foundation for training next-generation NIDS [10], [11]. Nevertheless, challenges remain. Chief among them are class imbalance, inefficient feature selection, and the generation of false positives, which can reduce detection reliability [12]. Addressing these challenges through preprocessing techniques—such as normalization, handling missing values, and balancing class distributions—is essential to building more accurate and responsive intrusion detection models. This paper contributes to this ongoing effort by performing a comparative analysis of five machine-learning algorithms using the CSE-CIC-IDS2018 dataset. The goal is to determine which models perform best in detecting various types of cyberattacks, specifically focusing on improving detection performance through effective data-balancing strategies.

Cyberattacks are becoming increasingly frequent and sophisticated, affecting critical sectors such as finance, healthcare, and government systems [12]. Machine learning models have shown great potential in detecting these threats, but significant challenges remain in improving their adaptability and efficiency [13]. These trends highlight the need for more robust and adaptive intrusion detection strategies. In order to address these challenges, this study utilizes the CSE-CIC-IDS2018 dataset, a comprehensive collection of real-world attack scenarios providing a solid foundation for building more effective intrusion detection systems (IDS) [15]. Key data preprocessing steps, such as cleaning, normalization, and feature extraction, are performed to enhance detection accuracy [16]. Various machine learning models are trained to identify unusual patterns that may indicate cyberattacks. The performance of these models is measured using key metrics like accuracy, precision, recall, and F1-score [17][67]. The algorithms compared include Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Logistic Regression, and XGBoost [18]. All experiments are conducted in a Jupyter Notebook environment using Python-based libraries for data processing, model training, and performance evaluation. This research aims to improve network intrusion detection by identifying the most effective methods for detecting and stopping cyber threats in network traffic [20].

Network Intrusion Detection Systems (NIDS) are a critical first line of defense in cybersecurity, designed to identify and respond to threats before they cause severe damage. Their effectiveness depends on two major factors: the quality of datasets used for training and the sophistication of detection algorithms [21]. However, as cyber threats evolve in complexity and frequency, traditional approaches relying on static, outdated datasets are proving less effective. Researchers now emphasize the importance of dynamic, real-world datasets that better reflect modern cyberattack strategies [22]. At the same time, advancements in machine learning (ML) and deep learning (DL) are reshaping how NIDS detects and analyzes threats. Researchers aim to develop more adaptive, high-precision intrusion detection systems by integrating advanced analytical methods. This section explores current trends and challenges in NIDS research, mainly focusing on adopting modern datasets and the role of advanced learning techniques in improving detection accuracy and system resilience [23].

For years, datasets like KDD Cup'99 and NSL-KDD were considered the gold standard for intrusion detection research. However, these datasets no longer reflect the complexity of today's network environments or the advanced techniques used in modern cyberattacks [24]. The KDD Cup'99 dataset, developed in the late 1990s, was a significant milestone. However, it lacks representative modern traffic behaviors and fails to capture emerging threats such as polymorphic malware and zero-day attacks [25]. NSL-KDD was introduced as an improvement over KDD Cup'99, and it addressed some of its predecessor's flaws by reducing redundancy and balancing attack types. However, it still struggles to represent the diversity and sophistication of contemporary cyber threats, making it less effective for training modern NIDS [26]. With traditional datasets proving inadequate, researchers have shifted focus to more dynamic, real-world datasets such as CSE-CIC-IDS2018. Unlike older collections, CSE-CIC-IDS2018 is built from actual network traffic. It includes a variety of attack types (e.g., DDoS, brute-force attacks, and botnets), making it far more suitable for developing robust intrusion detection models [27][28]. The growing reliance on these modern, adaptive datasets highlights the need for continuous updates in NIDS research to ensure that systems are reactive and proactive in addressing threats as they emerge.

Machine learning has become an essential tool for detecting anomalies in NIDS. Its effectiveness depends on the quality of the training data and the ability to select relevant features for the models. Over time, numerous ML models have been developed, each offering unique strengths in identifying and mitigating cyber threats. For example:

- **Random Forest (RF)** – Uses an ensemble of decision trees to enhance accuracy and robustness and thus is particularly useful for handling imbalanced datasets [26].
- **Support Vector Machine (SVM)** – Performs well on complex classification tasks but requires careful parameter tuning for optimal performance [27].
- **K-Nearest Neighbors (KNN)** – A simple and effective method for intrusion detection; however, its performance degrades with high-dimensional data [28].
- **Logistic Regression (LR)** is a highly interpretable model that struggles to capture complex attack behaviors [29].
- **XGBoost** – Known for efficiently handling imbalanced datasets, it is well-suited for modern cybersecurity challenges [30]. As cybersecurity threats grow, more advanced machine learning models evolve, aiming to provide faster and more adaptive intrusion detection capabilities.

Deep learning techniques have gained significant attention in NIDS research due to their ability to recognize complex attack patterns that traditional ML models may overlook. These models effectively analyze intricate network traffic behaviors and detect real-time anomalies. Notable examples include:

- Multilayer Perceptron (MLP) – Works well on structured datasets like NSL-KDD and UNSW-NB15, though it may require extensive training data for optimal learning [31].

- Convolutional Neural Networks (CNNs) – These are commonly used to analyze network traffic patterns, especially in modern datasets such as CSE-CIC-IDS2018 [32].
- Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) are designed for sequential data, making them highly effective in detecting anomalies that unfold over time [33]. Researchers increasingly explore hybrid approaches combining ML and DL techniques as DL advances to develop more resilient and adaptive NIDS solutions.

Despite considerable progress, NIDS still faces critical challenges that hinder its full potential. One major issue is the continued reliance on outdated datasets (e.g., KDD Cup'99 and NSL-KDD), which no longer capture the complexity of modern threats. This limits the real-world applicability of models trained on such data, as they fail to recognize emerging attack patterns and evolving network behaviors [34][38]. Another related challenge is the lack of dataset diversity: many studies focus on a narrow set of benchmark sources, overlooking richer and more representative datasets like CSE-CIC-IDS2018 that offer a realistic view of contemporary threats [35][40]. Expanding the range of datasets used in NIDS research is essential for improving detection accuracy and system adaptability. Additionally, while traditional ML models remain widely used, DL techniques are still underutilized in this field. Deep learning has the potential to enhance detection capabilities greatly, yet many approaches do not leverage its full power. Future research should explore more advanced neural network architectures and hybrid models that combine ML and DL methods for robust, adaptable intrusion detection systems [36][37][39]. By addressing these challenges, researchers can develop next-generation NIDS that are more resilient, adaptive, and capable of handling an ever-evolving threat landscape.

Researchers focus on several key areas for improvement to address the challenges outlined above and further enhance NIDS' effectiveness. Two of the most prominent are improving the quality of training data and refining learning strategies. In particular, recent works emphasize the adoption of up-to-date datasets for model training and mitigating class imbalance issues in these datasets to boost model performance. One crucial step in improving intrusion detection accuracy is to use modern, high-quality datasets that better reflect real-world threats. The CSE-CIC-IDS2018 dataset has emerged as a strong alternative to older datasets like KDD Cup'99 and NSL-KDD, which no longer capture the complexity of today's attacks. Researchers argue that continuing to rely on outdated datasets severely limits the practical effectiveness of NIDS [40]. In contrast, CSE-CIC-IDS2018 encompasses a broad range of contemporary attack types—such as DDoS, brute force, botnets, and even ransomware—making it more suitable for training advanced ML models [41][42]. Adopting such up-to-date datasets ensures that intrusion detection models are better equipped to handle emerging threats and adapt to evolving attack patterns. One major challenge associated with CSE-CIC-IDS2018 is class imbalance — certain attack types appear far less frequently than regular network traffic. This imbalance can skew model training, leading to higher false favorable rates and reduced overall accuracy [43]. To mitigate the issue, researchers apply data-balancing techniques such as oversampling the minority attack classes (to increase the representation of rare attacks) and undersampling the abundant regular traffic (to reduce its dominance in the training set). These methods help an intrusion detection model recognize common and rare threats, improving its overall performance [44]. If the class imbalance is ignored, ML models can become biased towards the majority class, making them unreliable at detecting low-frequency yet critical cyberattacks [45]. Therefore, addressing class imbalance is crucial for achieving optimal model accuracy and maintaining robust cybersecurity defenses

II. METHODOLOGY

This study follows a structured approach to improving machine learning models for intrusion detection, with a strong emphasis on correcting class imbalance. The process begins with thorough data preprocessing, which includes:

- Handling missing values to ensure data completeness.
- Normalizing features for consistency.
- Balancing the dataset to improve model reliability and accuracy.

After preprocessing, the dataset undergoes partitioning into two subsets for training and testing:

- 80% for training – allowing the models to learn patterns.
- 20% for testing – enabling fair evaluation of unseen data.

Machine learning algorithms are applied, including Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Logistic Regression, and XGBoost. Each model is assessed using key performance metrics such as accuracy, precision, recall, and F1-score [46]. These metrics provide valuable insights into how well each model detects intrusions and its ability to identify new, unseen attack patterns. The ultimate goal is to develop exact and adaptable intrusion detection models to tackle real-world cybersecurity threats [47].

This study utilizes the CSE-CIC-IDS2018 dataset (specifically the 02-15-2018 portion), which provides a diverse representation of regular network traffic and various cyberattack instances. It is one of the most comprehensive recent datasets for training and testing intrusion detection systems (IDS) [45]. The dataset comprises 80 features and approximately 1,548,575 data records, offering a rich basis for modeling network traffic. However, a significant challenge with this dataset is a class imbalance — regular traffic far outweighs intrusion events. This imbalance can hinder a model's ability to learn patterns of rare attacks [46]. If left unaddressed, the skewed class distribution may lead to a high false-negative rate (i.e., real threats going undetected). In order to mitigate this, data balancing techniques such as oversampling minority attack classes or undersampling the majority class are often applied. In our methodology, we employ an undersampling approach to represent the underrepresented attack traffic better, thereby improving overall detection accuracy [47]. By applying robust preprocessing and class-balancing strategies, the reliability and performance of the intrusion detection models can be significantly improved, making them more effective in real-world scenarios.

A summary of the data processing and modeling pipeline is illustrated in Fig. 1.

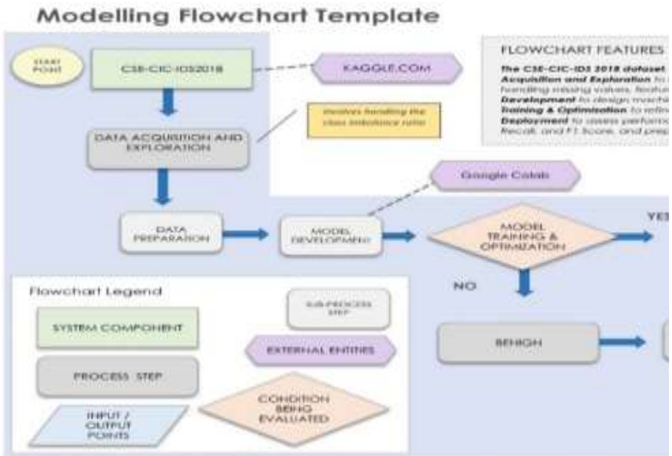

Fig. 1. Flowchart of the machine learning modeling process for network intrusion detection.

To address key challenges in the CSE-CIC-IDS2018 dataset—such as class imbalance, missing values, and data formatting inconsistencies—thorough data preprocessing was performed [48]. After the dataset was prepared correctly, machine learning models—including Random Forest, SVM, XGBoost, Logistic Regression, and KNN—were better positioned to achieve optimal performance while reducing the risk of overfitting. As a result, the overall reliability and accuracy of intrusion detection systems (IDS) improved. One widely used method for managing missing values was linear interpolation. This technique estimated missing entries by assuming a steady transition between known data points, ensuring the dataset remained consistent and dependable for training [49]. By reducing biases in the data, such interpolation helped improve model stability. A well-prepared dataset was essential for boosting model performance and ensuring that IDS operated effectively in real-world cybersecurity applications.

Missing values can significantly affect the accuracy and reliability of machine learning models, particularly in network intrusion detection [50]. Although various algorithms handle missing data differently, maintaining model integrity requires effectively addressing these gaps. In this study, linear interpolation was employed to construct a well-balanced dataset and minimize the negative impact of incomplete data on predictive performance. Accurate handling of missing values is critical for preserving model fidelity and supporting effective anomaly detection in network traffic.

Handling Missing Data Across Different Algorithms:

- Random Forest (RF) – Utilises built-in mechanisms to manage missing values, but additional preprocessing can further enhance accuracy [47].
- Support Vector Machines (SVM) – Require complete datasets, as missing values disrupt decision boundaries [48].
- XGBoost – Incorporates internal strategies for handling missing values, although external preprocessing improves stability and performance [48].
- Logistic Regression – Relies on linear relationships, necessitating imputation to maintain accurate probability estimation [49].
- K-Nearest Neighbors (KNN) – This is particularly sensitive to missing data due to its distance-based nature, requiring careful imputation to retain predictive reliability [49].

Employing effective data imputation techniques improves model performance, reduces bias, and enhances detection capability in Network Intrusion Detection Systems (NIDS) [50], [51].

A practical method for estimating missing values is linear interpolation. This technique works by drawing a straight line between two known data points and assuming that changes occur steadily between them. This method estimates missing values by connecting two surrounding data points in a straight line, preserving the underlying trend and maintaining consistency across structured datasets.

The formula for linear interpolation is:

$$y = y_1 + \frac{(x - x_1)(y_2 - y_1)}{x_2 - x_1} \tag{1}$$

This method is particularly effective when the data follows an approximately linear trend, as it ensures a smooth transition between values without introducing significant distortions [52]. In this study, the interpolate function from the pandas library was utilized to estimate missing values in numeric columns, allowing for seamless adjustments while preserving data integrity. This approach helps maintain continuity in data patterns, reducing the risk of bias during model training and improving overall consistency in the intrusion detection system [53].

Feature scaling ensures that all dataset features contribute equally to a model's predictions, preventing larger numerical values from dominating the learning process. This step is crucial in network intrusion detection, where varying feature magnitudes can distort model performance. Techniques like normalization and standardization help adjust feature values, ensuring that no single attribute overpowers decision-making [52]. Scaling is especially crucial for distance-based models such as Support Vector Machines (SVM) and K-nearest neighbors (KNN). These models can miscalculate distances between data points without proper scaling, leading to skewed predictions and reduced accuracy [53]. Standardizing feature values makes the dataset more uniform, improving model performance and ensuring fairer, more reliable classification.

Handling the imbalance between normal and attack traffic is critical for ensuring unbiased intrusion detection models. To address this issue, an undersampling technique was applied, reducing the number of normal traffic entries to match the number of attack instances. By balancing both classes, the model avoids favoring the majority class, leading to more accurate and fair predictions [56]. After undersampling, each category contained an equal number of instances, resulting in a more balanced training dataset. This adjustment prevents models from being skewed toward normal traffic, ensuring that intrusions are detected more effectively without unnecessary bias.

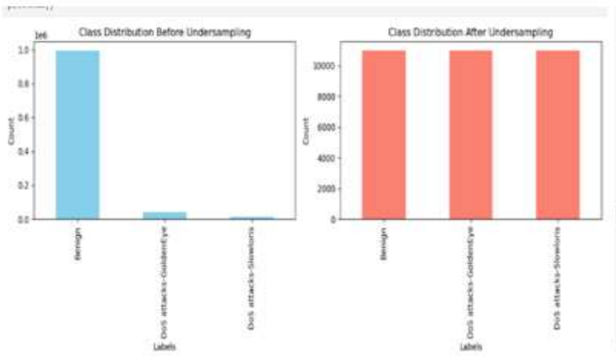

Fig. 2. Class Label Distribution -A Visual Comparison of Counts before and After Undersampling

This study explores multiple machine-learning algorithms for network intrusion detection. Each model follows a distinct approach to classifying network traffic, and their key mathematical formulations are presented below.

Random Forest is an ensemble learning method that constructs multiple decision trees, making predictions based on the majority vote among the trees. It is known for its robustness and resistance to overfitting, making it highly effective for detecting network intrusions [57].The prediction $\hat{y}$ from a Random Forest for classification is:

$$\hat{y} = mode\ (T_1(\mathrm{x}), T_2(\mathrm{x}), \ldots., T_n(\mathrm{x})) \qquad (2)$$

Where:

T_i (x) is the prediction of the i-th decision tree for input x

n is the total number of trees in the Random Forest model

Support Vector Machines (SVM) classify data by identifying the optimal hyperplane that separates different classes while maximizing the margin between them. When handling non-linearly separable data, kernel functions are used to transform the feature space, making SVM an effective choice for network intrusion detection [58].

The equation defining the decision boundary for SVM is:

$$w^T x + b = 0 \qquad (3)$$

Where:

w is the weight vector, x is the input vector, b is the bias.

K-Nearest Neighbors (KNN) is a non-parametric learning algorithm that classifies data points based on their proximity to the nearest k-labeled neighbors. It is beneficial for anomaly detection in network security [59].

The distance metric, commonly Euclidean distance, is calculated as: The distance metric (e.g., Euclidean distance) is used to determine proximity:

$$d(x, xi) = \sqrt{\sum_{j=1}^{n} (xj - xij)^2} \qquad (4)$$

Where:

x is the input data point.

xi is the i-th data point in the dataset.

n represents the number of features.

Logistic Regression is widely used in classification tasks as it estimates the probability of a class occurrence using the logistic function. Its simplicity and interpretability make it a popular choice for network intrusion detection [60]. The probability of a data point belonging to class y = 1 given input x is modeled as:

$$P(y = 1|x) = \frac{1}{1 + e^{-(w^T x + b)}} \qquad (5)$$

Where:

w is the weight vector

x is the input vector.

b is the bias term.

XGBoost is a gradient-boosting algorithm that builds multiple models sequentially, each learning from the errors of previous iterations. This technique is highly efficient for intrusion detection, as it can handle large datasets and prevent overfitting [61]. The model prediction for an input is computed as follows:

the model's prediction is the sum of predictions from all trees:

$$\hat{y} = \sum_{k=1}^{k} f_k\,(x) \qquad (6)$$

Where:

$f_k(x)$ is the prediction from the k-th tree.

K is the total number of trees.

The objective function combines the loss function $L(\hat{y}, y)$ with regularization terms to prevent overfitting:

$$L = \sum_{i=1}^{n} L(y_{i,} \hat{y}_i) + \sum_{k=1}^{k} \Omega(f_k) \quad (7)$$

Where Ω (f_k) is a regularization term on the complexity of the model.

Assessing intrusion detection models involves measuring how well they identify threats while minimizing errors. In this study, the strengths and weaknesses of each model were examined using four key performance metrics: accuracy, precision, recall, and F1-score. The dataset was divided into 80% training and 20% testing data, ensuring a fair and unbiased evaluation. This approach guarantees that models are tested on unseen data, providing a realistic measure of practical performance [62].

III. RESULTS AND DISCUSSION

A comprehensive evaluation of five machine learning models—Random Forest, SVM, KNN, Logistic Regression, and XGBoost—was conducted using the CSE-CIC-IDS2018 dataset. This dataset, rich in normal and malicious network traffic, was an ideal testbed for benchmarking the models. By employing four pivotal metrics (Accuracy, Precision, Recall, and F1-Score), the study captured a complete picture of each model's intrusion detection capability, ensuring that every nuance of performance was considered.

Table I. summarizes the comparative performance of the models. It presents the accuracy, precision, recall, and F1 score each classifier achieves. These metrics offer a quantitative assessment of how well each model detects network intrusions under the same conditions. Additionally, Fig. 3. visually represents these evaluation results, making it easier to observe performance differences across the models. This graphical comparison enhances interpretability, allowing a clearer understanding of which models are most effective in detecting intrusions.

TABLE I. Evaluation of the Machine Learning Models Implemented

Model	Accuracy	Precision	Recall	F1-Score
XGBoost	0.96	0.94	0.98	0.96
Random Forest	0.95	0.92	0.98	0.95
SVM	0.92	0.9	0.94	0.92
KNN	0.9	0.88	0.92	0.9
Logistic Regression	0.88	0.86	0.9	0.88

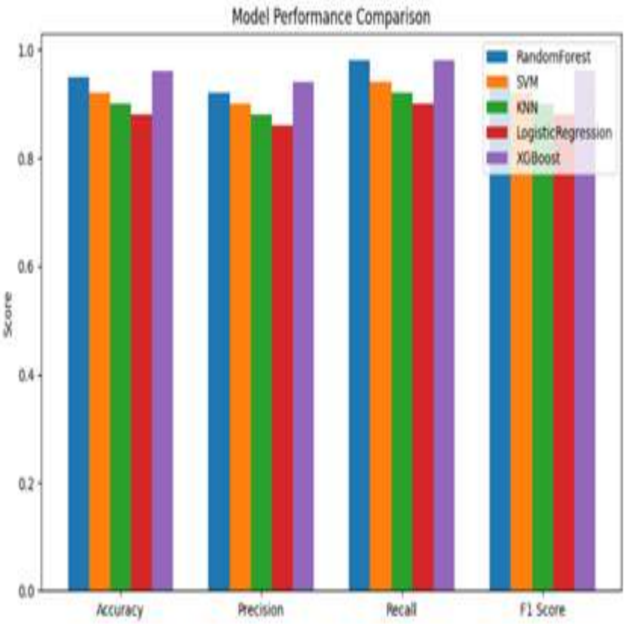

Fig. 3. Machine Learning Model Performance Comparison

XGBoost achieved the highest accuracy (96%) among the tested models, closely followed by Random Forest at 95%. This indicates that both models are highly effective at correctly classifying network traffic, making them particularly valuable for intrusion detection. SVM and KNN showed moderate accuracy levels (approximately 92% and 90%, respectively), reflecting a balance between correctly and incorrectly classified instances. Logistic Regression had the lowest accuracy (88%), suggesting that it struggled with the network data's complexity, adversely impacting its overall classification performance.

Precision is particularly critical in cybersecurity, as misclassifying benign traffic as malicious can lead to unnecessary alerts and system disruptions. In this aspect, XGBoost and Random Forest outperformed the others, achieving 94% and 92% precision rates, respectively. These high precision values indicate an ability to minimize false positives, thereby providing excellent reliability in distinguishing real threats from regular traffic. SVM achieved a precision of 90%, demonstrating solid performance in limiting false alarms. KNN followed with 88%, indicating slightly lower effectiveness. Logistic Regression had the lowest precision at 86%, reflecting a higher likelihood of misclassifying normal traffic as malicious and, thus, a greater risk of unwarranted security interventions.

The recall is crucial in cybersecurity applications, as it measures a model's ability to identify actual attacks without missing critical incidents. Higher recall values indicate that more real threats are detected, enabling security systems to respond effectively. In this study, the recall scores (summarized in Table II) illustrate each model's effectiveness in detecting malicious activities, reinforcing the overall evaluation of their intrusion detection capabilities.

TABLE II. Evaluation of Recall on Each Model Developed

Model	Recall	Effectiveness in Attack Detection
XGBoost	0.98	Highly effective
Random Forest	0.98	Highly effective
SVM	0.94	Captures significant attacks
KNN	0.92	Captures moderate attacks
Logistic Regression	0.9	Lower effectiveness

The F1 Score is a comprehensive metric that balances precision and recall, providing a single measure of a model's overall effectiveness in intrusion detection tasks. Among the evaluated models, XGBoost achieved the highest F1 Score of 0.96, demonstrating its strong ability to identify threats and minimize false alarms. Random Forest followed closely with an F1 Score of 0.95, further confirming its reliable classification performance. SVM and KNN produced moderate F1 Scores of 0.92 and 0.90, respectively, indicating that while they are reasonably effective, they are not as robust as XGBoost or Random Forest in balancing precision and recall. Logistic Regression, with the lowest F1 Score of 0.88, proved to be the least effective, highlighting its limitations in handling complex, high-dimensional network traffic patterns.

The comparative analysis of these machine learning models provides several key insights regarding their suitability for intrusion detection systems:

- **XGBoost**: This model stands out as the top performer, leading across accuracy, precision, recall, and F1-score. Its ability to minimize false positives while also detecting nearly all attacks makes it a strong candidate for deployment in real-world IDS implementations.
- **Random Forest**: Performing almost as well as XGBoost, Random Forest offers robust classification with the advantage of interpretability. It is a viable alternative in scenarios where explainability and reliability are as important as raw performance.
- **SVM and KNN**: These models demonstrate moderate effectiveness. While they may be suitable for lower-risk environments or as complementary tools, their computational complexity (especially for KNN) and lower overall performance make them less ideal as standalone solutions.
- **Logistic Regression**: This model's weakest performance indicates limited suitability for complex, high-dimensional network data. Its inability to distinguish intricate attack patterns effectively makes it suboptimal for modern intrusion detection tasks.

Overall, these findings underscore the value of advanced machine learning approaches—particularly XGBoost and Random Forest—in enhancing next-generation intrusion detection systems' accuracy, adaptability, and robustness.

IV. CONCLUSION AND RECOMMENDATIONS

This study conducted a comparative analysis of five machine learning algorithms—Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Logistic Regression, and XGBoost—using the CSE-CIC-IDS2018 dataset for network intrusion detection. Key preprocessing steps, including data cleaning, feature scaling, and class balancing through undersampling, were implemented to enhance model training and accuracy. Among the evaluated models, XGBoost emerged as the top performer across all four performance metrics: accuracy, precision, recall, and F1-score. Random Forest also delivered strong results, showing near-equivalent performance. These two models demonstrated superior ability to identify complex attack patterns while maintaining low false-positive rates, making them highly suitable for deployment in real-world intrusion detection systems. In contrast, SVM and KNN offered moderate performance, and Logistic Regression was the least effective, particularly in handling high-dimensional and imbalanced data.

Recommendations from this study include:

- Organizations implementing intrusion detection systems should prioritize ensemble learning models such as XGBoost or Random Forest due to their high reliability and adaptability.
- Data preprocessing—including proper handling of missing values, feature scaling, and class balancing—should be treated as essential for improving model performance and generalization.
- Future research could explore hybrid models that combine machine learning with deep learning techniques, as well as real-time deployment scenarios to evaluate practical system latency and adaptability.

This study reinforces the critical role of modern machine learning techniques and balanced datasets in enhancing cybersecurity and suggests promising directions for the development of next-generation intelligent intrusion detection systems.

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